

Supplementary Appendix

Ashkenazi Population Structure Reassessed: Reference-Panel Sensitivity and the Europe-Levant Cline

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Scope

This appendix preserves the limited pairwise FST outputs retained in the manuscript as descriptive secondary comparisons. The values were obtained through the IllustrativeDNA Admix Lab web interface using Human Origins v62.0 with Adjust Pseudohaploids set to NO. The underlying genotype files and the complete web-platform filtering pipeline were not available for independent audit. These tables are therefore not treated as primary evidence, formal demographic estimates, or a command-line replication.

Table S1. Pairwise FST distances from Jew_Ashkenazi.HO to declared comparator populations.

Rank	Comparator population	FST distance	Standard error
1	Cretan.DG	0.0049	0.00124
2	Italian_South.HO	0.0049	0.00033
3	Sicilian.HO	0.0052	0.00041
4	Greek.HO	0.0055	0.00035
5	Maltese.HO	0.0066	0.00045
6	Jew_Moroccan.HO	0.0091	0.00058
7	Jew_Iraqi.DG	0.0122	0.00138
8	Jew_Iranian.HO	0.0125	0.0005
9	Jew_Tunisian.HO	0.0127	0.00056
10	Jew_Libyan.HO	0.0132	0.00053
11	Jew_Georgian.HO	0.0133	0.00056
12	Jew_Yemenite.HO	0.0174	0.00054

Note. Distances are copied from the retained web-platform output. Similar numerical values do not identify a unique historical source, and the table does not provide confidence intervals for a demographic model.

Table S2. FST calibration comparisons retained for scale context.

Population 1	Population 2	FST distance	Interpretive context
Italian_South.HO	Maltese.HO	0.0036	Closely related central Mediterranean populations
Jew_Ashkenazi.HO	Italian_South.HO	0.0049	Strong regional affinity
IBS_CanaryIslands.DG	Spanish.HO	0.0038	Iberian-derived population with additional North African ancestry

Note. The Canary Islands comparison is included only to illustrate the scale of a low FST value between closely related populations with documented additional ancestry. It is not an ancestry model for Ashkenazi Jews.

Files and validation

Machine-readable copies are provided as supplementary/tables/Table_S1_FST_results.csv and supplementary/tables/Table_S2_FST_calibration.csv. The exact labels, comparison specifications, copied web output, screenshot, and validation script are in descriptive_fst/. Run python3 descriptive_fst/scripts/validate_fst_tables.py from the repository root to cross-check the tables against the retained records.